



CS435 Introduction to Big Data

Unit 2. In-Memory Distributed Computing Paradigm

Week 1-B

Introduction to In-Memory Distributed Computing Framework

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FAQs

- If you need to use a laptop, please sit in the last row
 - You can still use a tablet if you don't block any student behind you
- Do we have a lab session this week?
 - No.
 - The first lab sessions will be in the Week 2. 11AM-noon on Monday, 5-6PM on Tuesday, or 11AM-noon on Saturday (801 only).
 - Sign-up for your lab session, if you have not done yet.
- **[Section 801 ONLY]** MYP (Meet Your Professor) sessions are starting this Friday. If you have not signed up for it yet, please use the link included in the Canvas announcement. If there no time slot working for you, please let me know.



FAQs

- Where can I find the course information and video clips?
 - Canvas Modules!
 - All the “*static*” materials are there
 - Go to the Echo360 tab for the video clips
- Quizzes will be open-note. Only lecture notes/your personal notes are allowed.
- What if you need more help than the 24 hour setting itself?



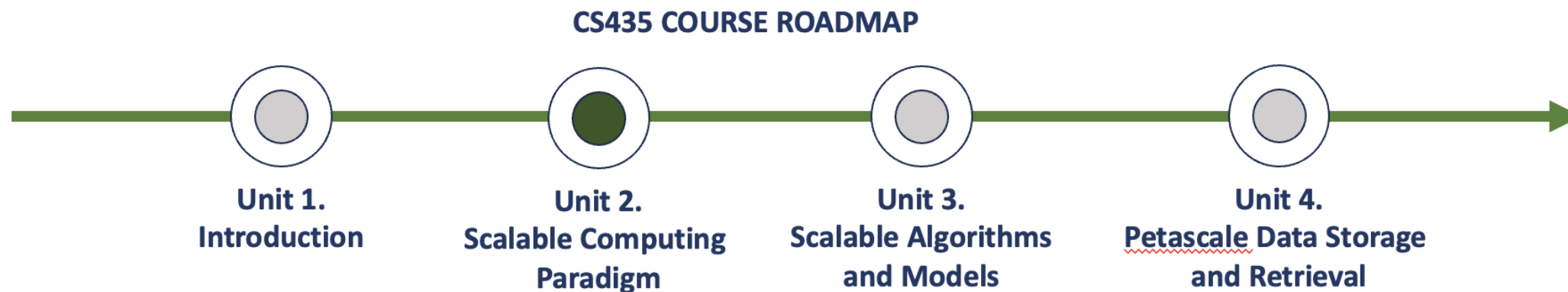
Topics covered in the previous class

- Introduction to Big Data
- Course introduction



Topics to be covered in today's class

- Overview of Big Data Analytics
- Introduction to MapReduce





Unit 2. Scalable Computing Paradigm

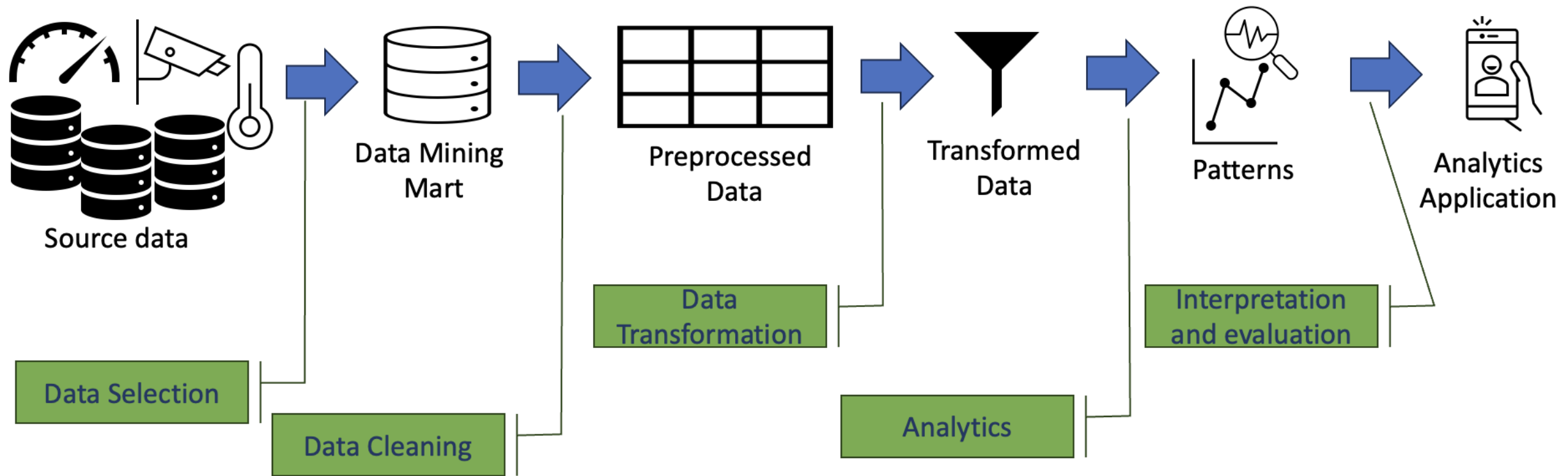
Overview of Big Data Analytics



This Material is Built Based on,

- Analytics in a Big Data World: The Essential Guide to Data Science and Its Applications, Bart Baesens, 2014, Wiley

Analytics Process Model





Types of Analytics

- Analytics is a term that is often used interchangeably with
 - Data science
 - Data mining
 - Knowledge discovery
- **Descriptive analytics**
 - No target variable
 - E.g., Clustering, association rules
- **Predictive analytics**
 - A target variable is typically available
 - E.g., linear/logistic regression, decision trees, neural networks, support vector machines



Types of Data Sources

- **Transactions**

- Structured, low-level, detailed information
- Customer transactions
 - Purchase, claim, cash transfer, credit card payment
- Stored in massive online transaction processing (OLTP) relational database
- Can be summarized over longer time horizons (e.g., averages, relative trends, Max/Min values)

- **Unstructured data embedded in text documents**

- emails, web pages, claim forms, audio and video, images
- Requires extensive preprocessing

- **Qualitative, expert-based data**

- Requires subject matter experts' (SME) analysis
- Scientific data

Sampling

- Taking a **subset** of data for analytics
 - Generating **hypothesis**
 - **Model** selection
 - **Feature** selection
 - Speculative process
 - Building **analytics model**
- **Stratified sampling**
 - Taking samples according to **predefined strata**
 - e.g., Fraud detection with very skewed (99 percent non-fraud customers, 1 percent fraud customers)
 - Sample should contain the same percentage of fraud customers as in the original data





Types of Data Elements

- **Numerical**

- **Continuous**

- Data elements that are defined on an interval that can be limited or unlimited
 - e.g., income, sales, temperature

- **Discrete**

- Typically counts
 - e.g., number of pages in a book

- **Categorical**

- **Nominal**

- Data elements that can only take on a limited set of values with no meaningful ordering between them
 - e.g., profession code (e.g., A..Z), purpose of loan (e.g., 1..10)

- **Ordinal**

- Data elements that can only take on a limited set of values with a meaningful ordering between them
 - e.g., credit rating groups (low, middle, high), age coded as young, middle age and old)

- **Binary**

- Data elements that can only take on two values
 - e.g., Having child, allowed to drive

Missing Values

- What will cause missing values?



Missing Values

- **Missing values** can occur because of various reasons
 - The information can be **non-applicable**
 - e.g., Survey questions
 - The information can be **undisclosed**
 - e.g., security or privacy sensitive information
 - The information can be **unavailable**
 - e.g., sensors/device/network issues



Missing Values --continued

- **Replace (impute)**

- Replaces the missing value with a **computed/selected value**
- Imputation algorithm examples
 - **Hot-deck**: replaces with a randomly selected similar records
 - **Cold-deck**: selects replacement from another dataset
 - **Mean substitution**: replaces with the mean of that variable for all other cases
 - **Regression**: predicts missing values of a variable based on other variables.



- **Delete**

- Deletes observations with lots of missing values
- This assumes that information is missing at random and has no meaningful interpretation and/or relationship to the target

- **Keep**

- Missing values can be meaningful
 - e.g., a customer did not disclose the income for current condition

Input



Raw images with cloud mask

Predicted

Cloud removed image
using CloudNet

Target



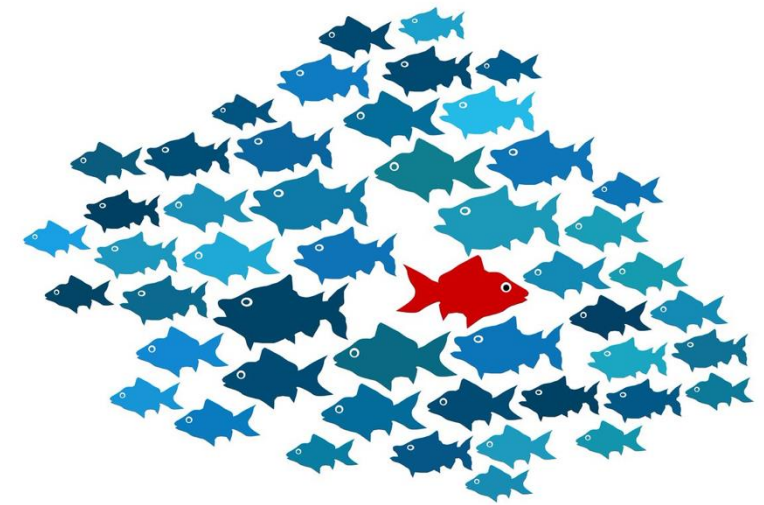
True image without cloud

Pixels covered with cloud: Missing data because it's unavailable.
Data was **imputed** using a deep learning model.

Khandelwal, Paahuni, Samuel Armstrong, Abdul Matin, Shrideep Pallickara, and Sangmi Lee Pallickara. "CloudNet: A Deep Learning Approach for Mitigating Occlusions in Landsat-8 Imagery using Data Coalescence." In *2022 IEEE 18th International Conference on e-Science (e-Science)*, pp. 117-127. IEEE, 2022.

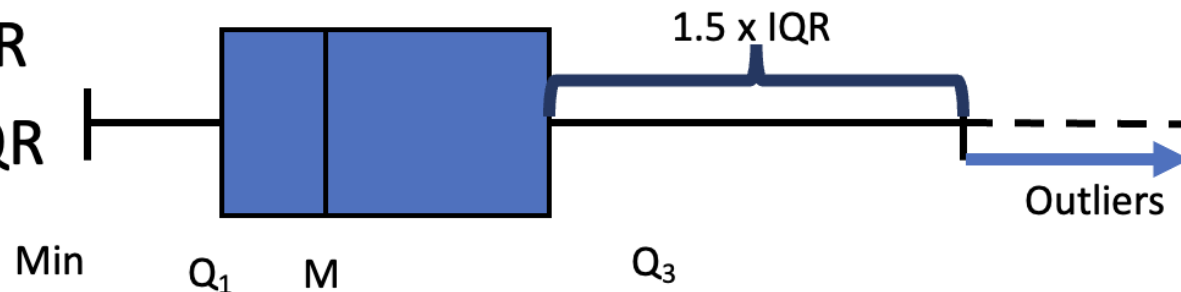
Outliers of Dataset

- **Outliers** are extreme observations that are very dissimilar to the rest of the population
 - **Valid observation**
 - Salary of boss
 - **Invalid observation**
 - Age is 300
- **Multivariate outliers**
 - Observations that are outlying when the multiple dimensions are combined
 - e.g., Temperature in Fort Collins is 100 degrees but on a midnight in December



Identifying Outliers using Box Plots

- A box plot represents three key quartiles of the data
 - Q_1 : 25% of the observations have a lower value
 - Q_2 : 50% of the observations have a lower value
 - Q_3 : 75% of the observations have a lower value
 - The **minimum** and **maximum** values are added
- **Too far away** is now quantified as more than **1.5 x Interquartile Range**
($IQR = (Q_3 - Q_1)$)
- Lower fence: $Q_1 - 1.5 \times IQR$
- Upper fence: $Q_3 + 1.5 \times IQR$



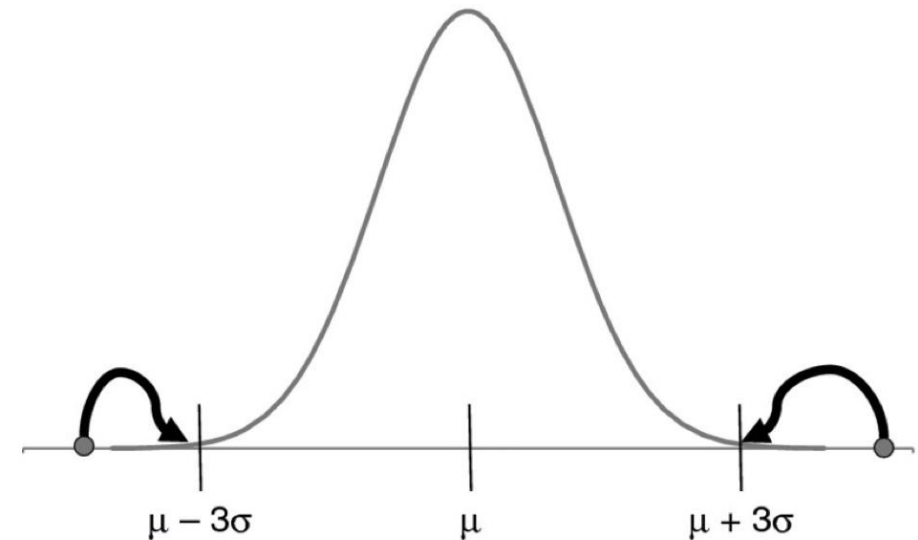
Identifying Outliers using Z-Score

- Measuring how many standard deviations an observation is away from the mean
 - $$Z_i = \frac{x_i - \mu}{\sigma}$$
where μ represents the **average of the variable** and σ its **standard deviation**
- A practical rule of thumb then defines outliers when the absolute value of the z-score $|z|$ is bigger than 3

ID	Age	Z-Score
1	30	$(30-40)/10=-1$
2	50	$(50-40)/10=+1$
3	10	$(10-40)/10=-3$
4	40	$(40-40)/10=0$
5	60	$(60-40)/10=+2$
6	80	$(80-40)/10=+4$
--	..	...
	$\mu = 40$ $\sigma = 10$	$\mu = 0$ $\sigma = 1$

Dealing with Outliers

- Treat outliers as **missing values**
- Popular schemes
 - **Truncation**
 - Taking only values that are within the limits
 - **Winsorizing**
 - Limiting extreme values to reduce the effect of possible spurious outliers
 - {92, 19, **101**, 58, **1053**, 91, 26, 78, 10, 13, **-40**, **101**, 86, 85, 15, 89, 89, 28, **-5**, 41} (N = 20, mean = 101.5)
→ {92, 19, **101**, 58, **101**, 91, 26, 78, 10, 13, **-5**, **101**, 86, 85, 15, 89, 89, 28, **-5**, 41} (N = 20, mean = 55.65)



Using the Z-Scores for truncation



Standardizing Data

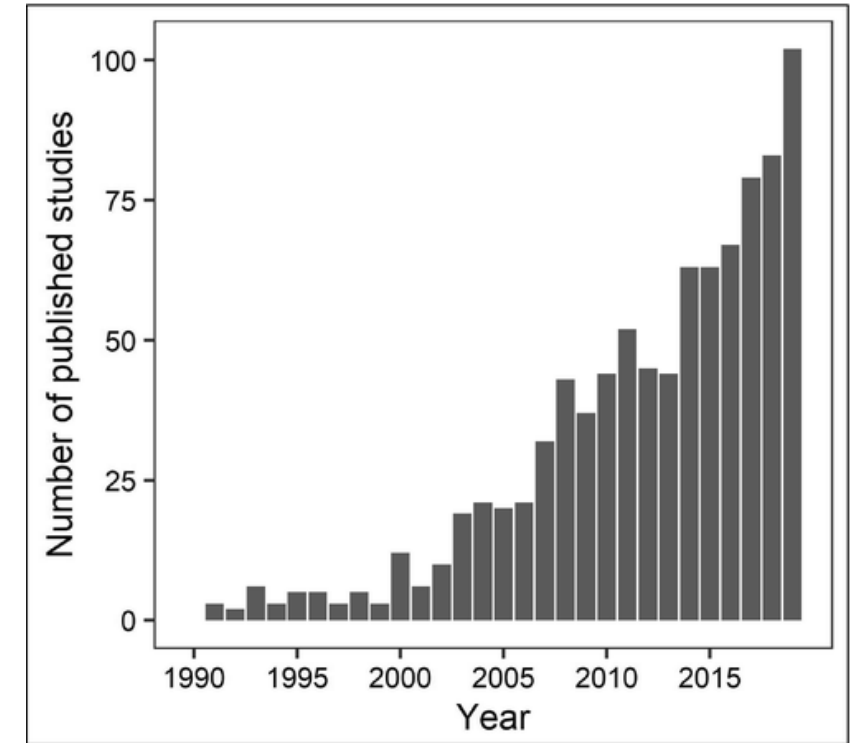
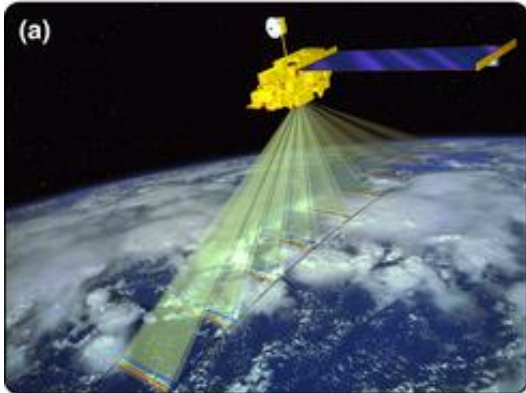
- Scaling variables to a similar range

- Min/Max standardization

- $$X_{new} = \frac{X_{old} - \min(X_{old})}{\max(X_{old}) - \min(X_{old})} (newmax - newmin) + newmin$$
- Where *newmax* and *newmin* are the newly imposed *maximum* and *minimum* (e.g. 1 and 0)

Data Integration

- Combining data from two or more disparate sources into one single source
- Challenges
 - Disparate data formats, sources, and entity resolutions
 - Poor data availability for some integrated data
 - Calibrating misaligned datasets
 - Data privacy
 - Imbalanced data – uneven quantities



Number of peer-reviewed publication that
Use of data integration in ecological research

Zipkin, Elise F., Erin R. Zylstra, Alexander D. Wright, Sarah P. Saunders, Andrew O. Finley, Michael C. Dietze, Malcolm S. Itter, and Morgan W. Tingley. "Addressing data integration challenges to link ecological processes across scales." *Frontiers in Ecology and the Environment* 19, no. 1 (2021): 30-38.

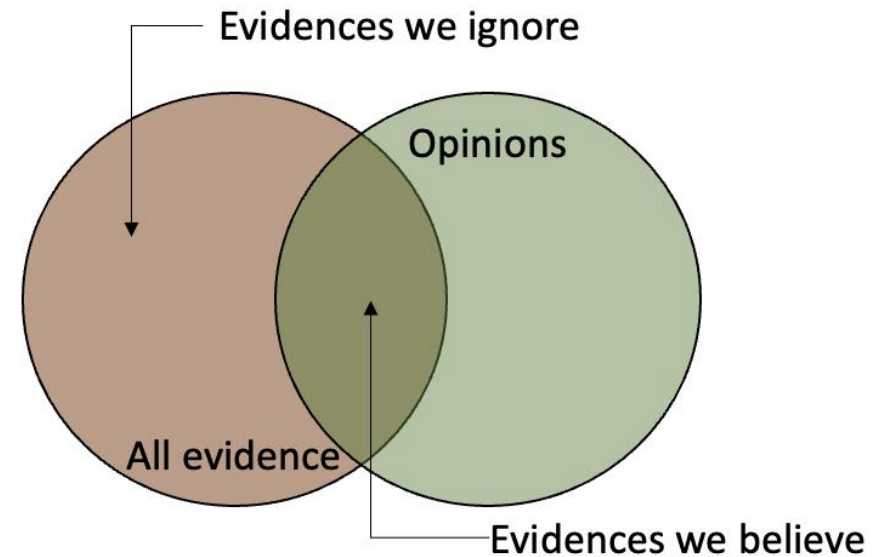
Common Biases in Big Data

- **Confirmation bias**

- Analysts tend to lean towards data that is in alignment with their opinions

- **Availability heuristic (a.k.a. availability bias)**

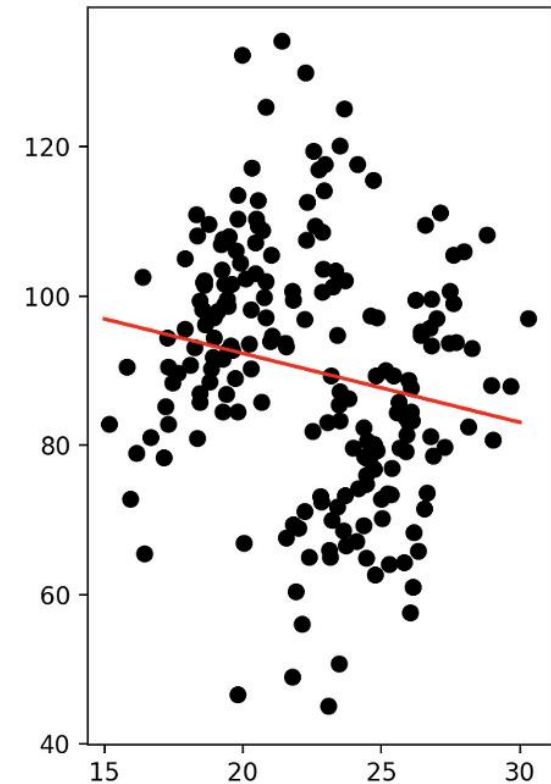
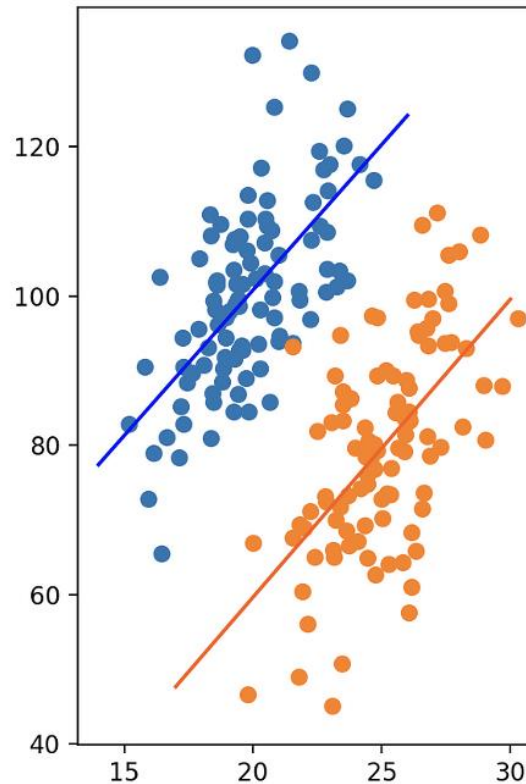
- Analysts make inferences based on readily available data or recent information alone



Common Biases in Big Data

- **Simpson's paradox**

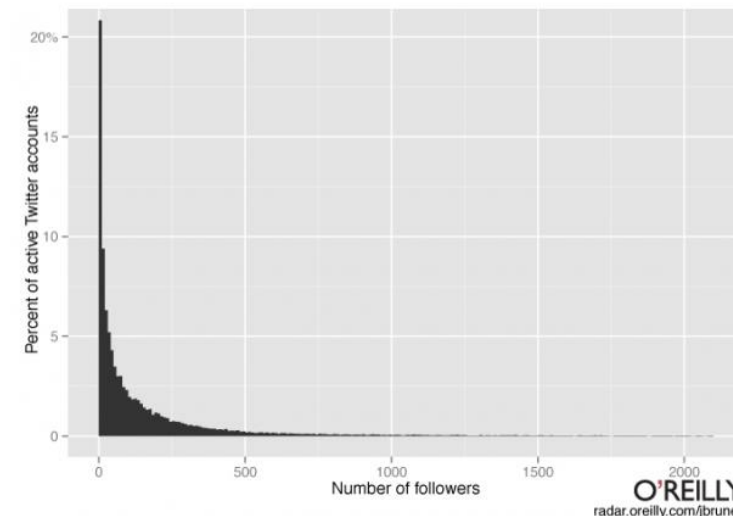
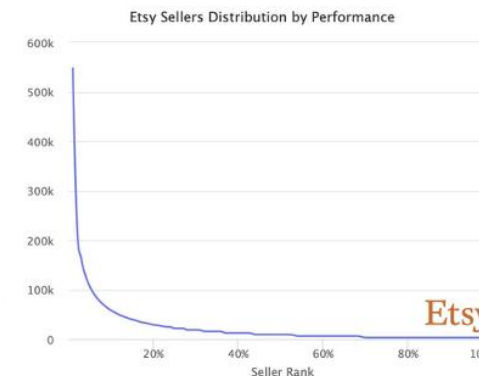
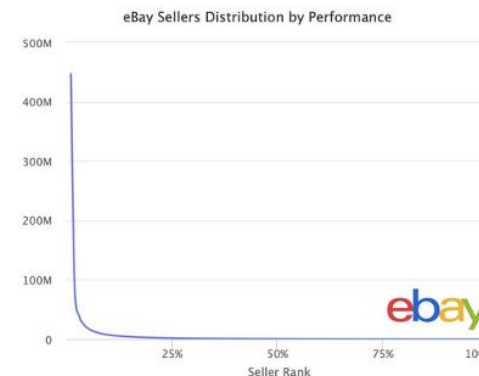
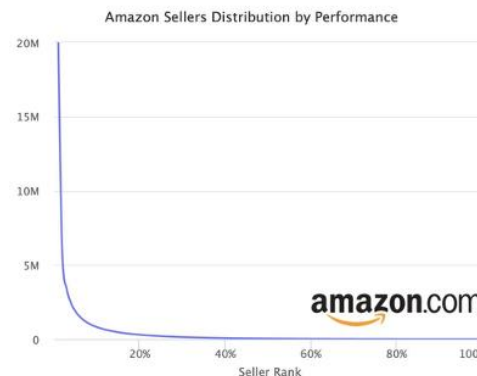
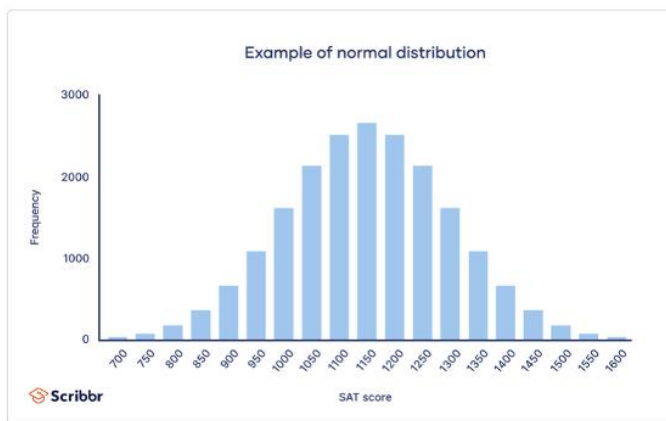
- Disparity between a pattern when analyzed in individual groups vs. cumulative group



Common Biases in Big Data

- Non-normality bias**

- Analysts assume that the aggregated data has the normal distribution

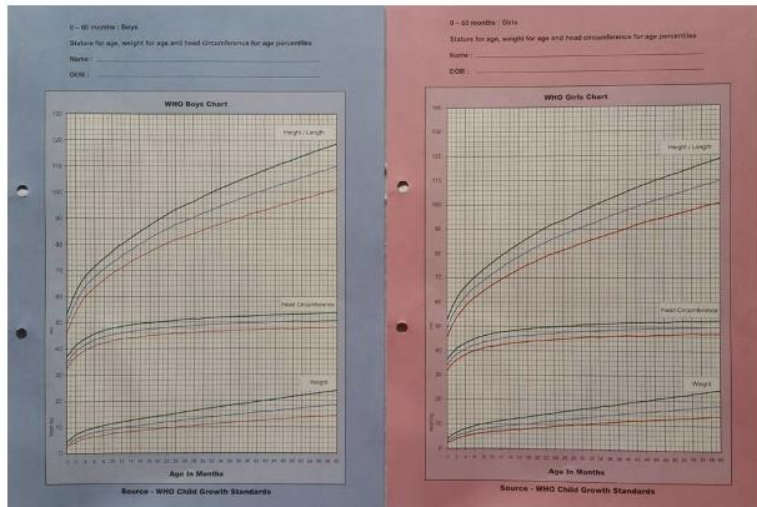


$p(x) \propto x^{-\alpha}$, where α is the exponent or scaling parameter

Common Biases in Big Data

• Overfitting and Underfitting

- Analysts assume that an overly complex model that encompasses several data trends is bound to bring accurate inferences

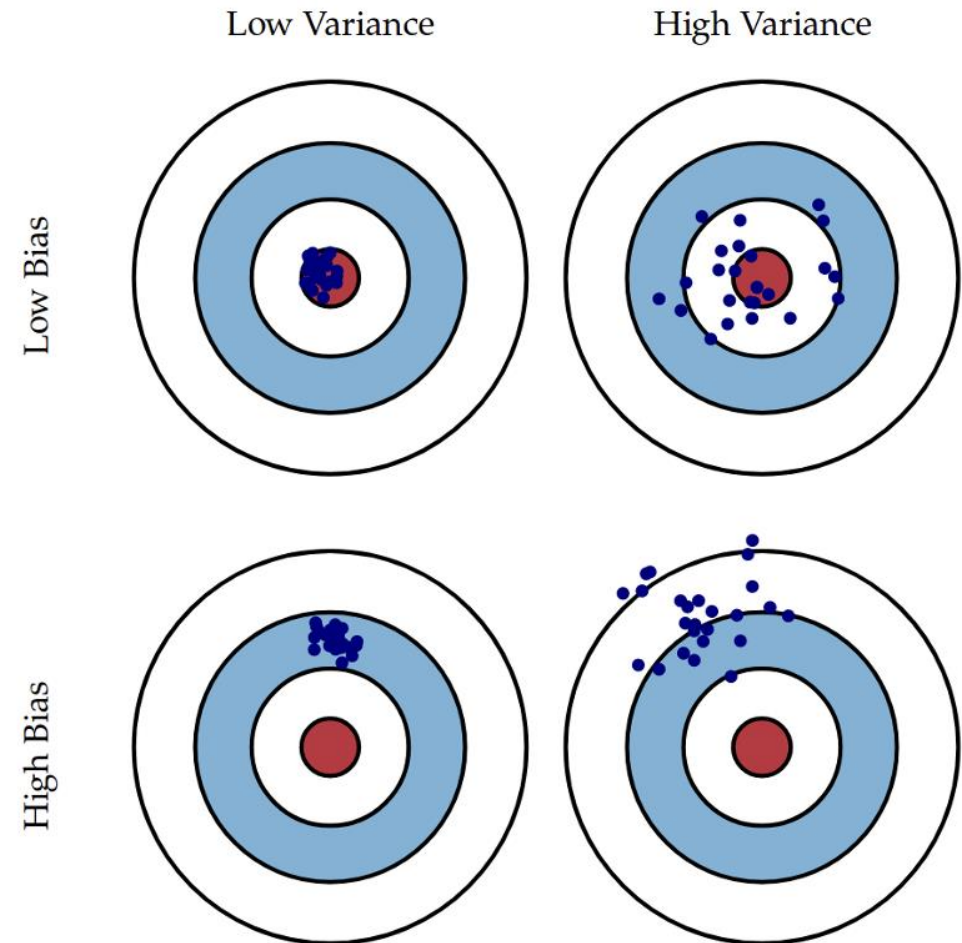


Boy's Chart (0-5 years)

© Gauri Kshirsagar

Girl's Chart (0-5 years)

www.raisingtwinsblog.wordpress.com





Unit 2. Scalable Computing Paradigm

Introduction to In-Memory Distributed Computing Framework



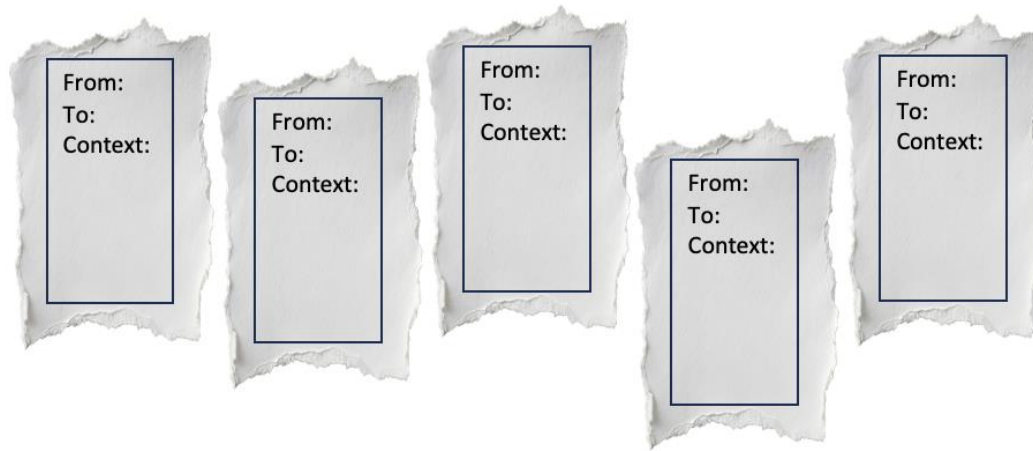
This material is built based on

- Matei Zaharia, Mosharaf Chowdhury, Tathagata Das, Ankur Dave, Justin Ma, Murphy McCauley, Michael J. Franklin, Scott Shenker, and Ion Stoica, “*Resilient Distributed Datasets: A Fault-Tolerant Abstraction for In-Memory Cluster Computing*,” The 9th USENIX Symposium on Networked Systems Design and Implementation (NSDI 12)
- Holden Karau, Andy Komwinski, Patrick Wendell and Matei Zaharia, “Learning Spark”, O’Reilly, 2015
- Spark programming guide
 - <https://spark.apache.org/docs/latest/>
 - Job Scheduling
 - <https://spark.apache.org/docs/2.0.0-preview/job-scheduling.html>



Worksheet

1. Form a small group of 4–5 students who are sitting in the same row as you.
2. Cut out the five attached communication sheets. Next, write 30 integers on the back of this sheet.



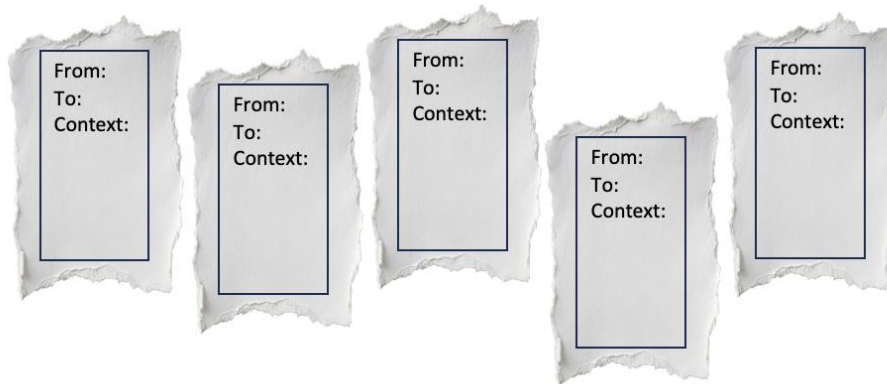
{3, 13, 23, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 43, 53, 63, 73, 83, 93,
103, 113, 123, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 143,
153, 163, 173, 183, 193, 203, 213, 223, 230, 231, 232, 233, 234, 235,



Worksheet

3. Your group's task is: "Calculate the sum of all the numbers written by your group members."
4. Assume that you are not allowed to share your numbers directly. However, you may communicate using the five communication sheets. How will you work with your local set of numbers while collaborating with other group members?
5. Write pseudocode describing your approach.

Requirement: All group members must run the same code to complete the task



```
{3, 13, 23, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 43, 53, 63, 73, 83, 93,  
103, 113, 123, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 143,  
153, 163, 173, 183, 193, 203, 213, 223, 230, 231, 232, 233, 234, 235,
```




Worksheet



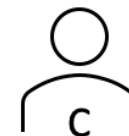
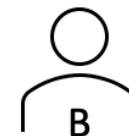
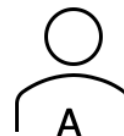
Write a pseudocode that each of your group members will execute. The same code will be representing the local task performed by each member.

How many of the communication sheets did your group use to complete this computation?

i.e., All members must run the same code.

Each of us will aggregate the data points
 Pass it to the person sitting in the righthand side
 Repeat this until there is no more data available
 from the person sitting in the righthand side
 If all the data is collected, calculate the global
 sum and report the value.

```
{3, 13, 23, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 43, 53, 63, 73, 83, 93,
103, 113, 123, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 143,
153, 163, 173, 183, 193, 203, 213, 223, 230, 231, 232, 233, 234, 235,
236, 237, 238, 239, 243, 253, 263, 273, 283, 293, 300, 301, 302, 303,
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332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345,
346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359,
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633, 634, 635, 636, 637, 638, 639, 643, 653, 663, 673, 683, 693, 703,
713, 723, 730, 731, 732, 733, 734, 735, 736, 737, 738, 739, 743, 753,
763, 773, 783, 793, 803, 813, 823, 830, 831, 832, 833, 834, 835, 836,
837, 838, 839, 843, 853, 863, 873, 883, 893, 903, 913, 923, 930, 931,
932, 933, 934, 935, 936, 937, 938, 939, 943, 953, 963, 973, 983, 993}
```

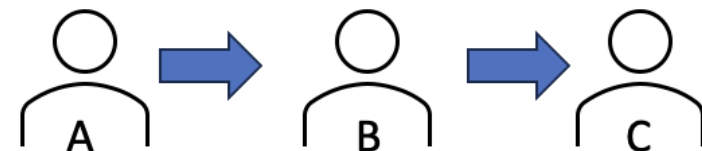




```

1: my_sum = 0; sum_left = 0
2: I_am_rightmost = findOut(my_location)
3: my_sum = calculate the local sum of my portion of
  the data
4: while (my_sum != -1)
5:     sum_left = retrieve the sum from the left student
6:     if (i_am_rightmost) {
7:         if (sum_left = null) {
8:             report my_sum
9:             my_sum = -1
10:        }
11:        else my_sum = my_sum + sum_left
12:    }
13: else {
14:     if (sum_left = null) my_sum = -1
15:     else my_sum = sum_left
16: }
17: }

```



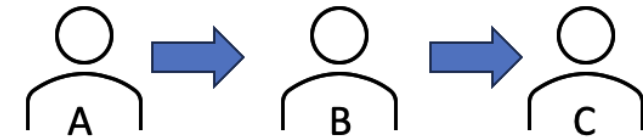
Variables	A	B	C
<u>my_sum</u>	0	0	0
<u>i_am_rightmost</u>	0	0	1
<u>my_sum</u>	203	94	12
<u>sum_left</u>	null	203	94
<u>my_sum</u>	-1	203	106
<u>sum_left</u>	--	null	203
<u>my_sum</u>	--	-1	309
<u>sum_left</u>	--	--	null
<u>Final result = 403</u>	--	--	-1

Will this work?



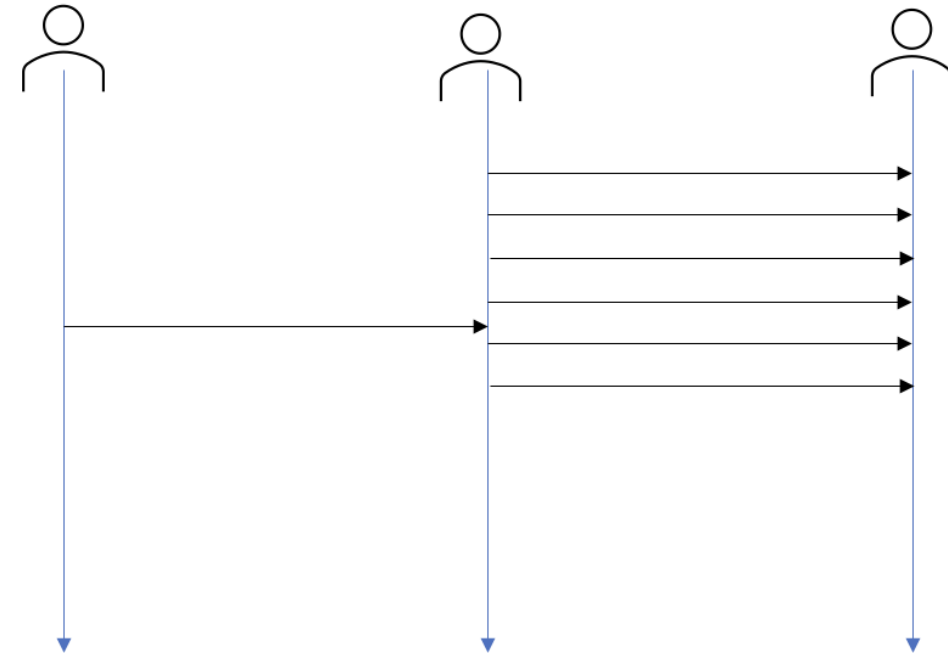


```
1: my_sum = 0; sum_left = 0
2: i am rightmost = findOut(my location)
3: my_sum = calculate the local sum of my portion of the data
4: while (my_sum != -1)
5:     sum_left = retrieve the my_sum from the left student
6:     if (i am rightmost) {
7:         if (sum_left = null) {
8:             report my_sum
9:             my_sum = -1
10:        }
11:        else my_sum = my_sum + sum_left
12:    }
13: else {
14:     if (sum_left = null) my_sum = -1
15:     else my_sum = sum_left
16: }
17: }
```



How can we get the i am rightmost variable?
What if my_sum is not ready for the student A and the student B tries to retrieve it? Is it **null**?
How does this algorithm indicate all the participants completed their computing and passed the value successfully?
What if the student B cannot finish the computing?

-
-
-





Computing frameworks for the distributed computing environments

- **Communication mode**
 - Synchronous vs. Asynchronous
- **Parallelization aspect**
 - Implementation of parallelization method
 - Model, data, or hybrid parallelization
- **Supporting device**
 - CPU, GPU, TPU
- **Supported architecture**
 - Centralized or decentralized architecture
- **Class of target tasks**
 - Machine learning, visualization
- Fault tolerance

Different frameworks are specialized with different tasks

Table 3

Distributed Machine Learning Framework comparison according to the first four classification criteria.

Framework	Parallelization Mode		Supporting Device	Supported Architecture		Communication Mode	
	Model Parallelism	Data Parallelism		Centralized	Decentralized	Asynchronous	Synchronous
H2O	X	X	CPU, GPU	X	X	X	
Vertica		X	CPU	X	NA		X
PyCOMPSs - dsLib		X	CPU, GPU	X	X	X	X
Cymbalo		X	CPU	X		X	X
HotML	X	X	CPU	X			X
Litz		X	CPU, GPU	X	X	X	X
SIREN	X	X	CPU		X	X	X
Apache Spark		X	CPU, GPU	X		X	X
KungFu	X	X	CPU, GPU	X	X	X	



Table 5
Distributed Deep Learning Framework comparison according to classification criteria.

Framework	Parallelization Mode		Supporting Device	Supported Architecture		Communication Mode	
	Model Parallelism	Data Parallelism		Centralized	Decentralized	Asynchronous	Synchronous
PyTorch	X	X	CPU, GPU, TPU	X	X	X	X
TensorFlow	X	X	CPU, GPU, TPU	X	X	X	X
Caffe2		X	CPU, GPU		X		X
Chainer	X	X	CPU, GPU, TPU		X		X
BigDL2.0		X	CPU, GPU	X	X		X
Elephas		X	CPU, GPU	X		X	X
SINGA	X	X	CPU, GPU	X	X	X	X
TensorFlowOnSpark	X	X	CPU, GPU	X	X	X	X
OneFlow	X	X	CPU, GPU		X		X
Horovod	X	X	CPU, GPU	X	X	X	X
SDDLf	X	X	CPU	X		X	
TFSM		X	CPU, GPU	X		X	
ShmCaffe		X	CPU, GPU	X		X	X
Petuumv2	X	X	CPU, GPU	X	X	X	X
MXNet	X	X	CPU, GPU	X	X	X	X
Petastorm	X	X	CPU, GPU	X	X	X	X
DeepSpeed	X	X	CPU, GPU	X	X	X	X
DeepSpark		X	CPU, GPU	X		X	
CoCoNet	X	X	CPU, GPU		X	X	X

Berloco, Francesco, Vitoantonio Bevilacqua, and Simona Colucci. "Distributed analytics for big data: A survey." *Neurocomputing* 574 (2024): 127258.

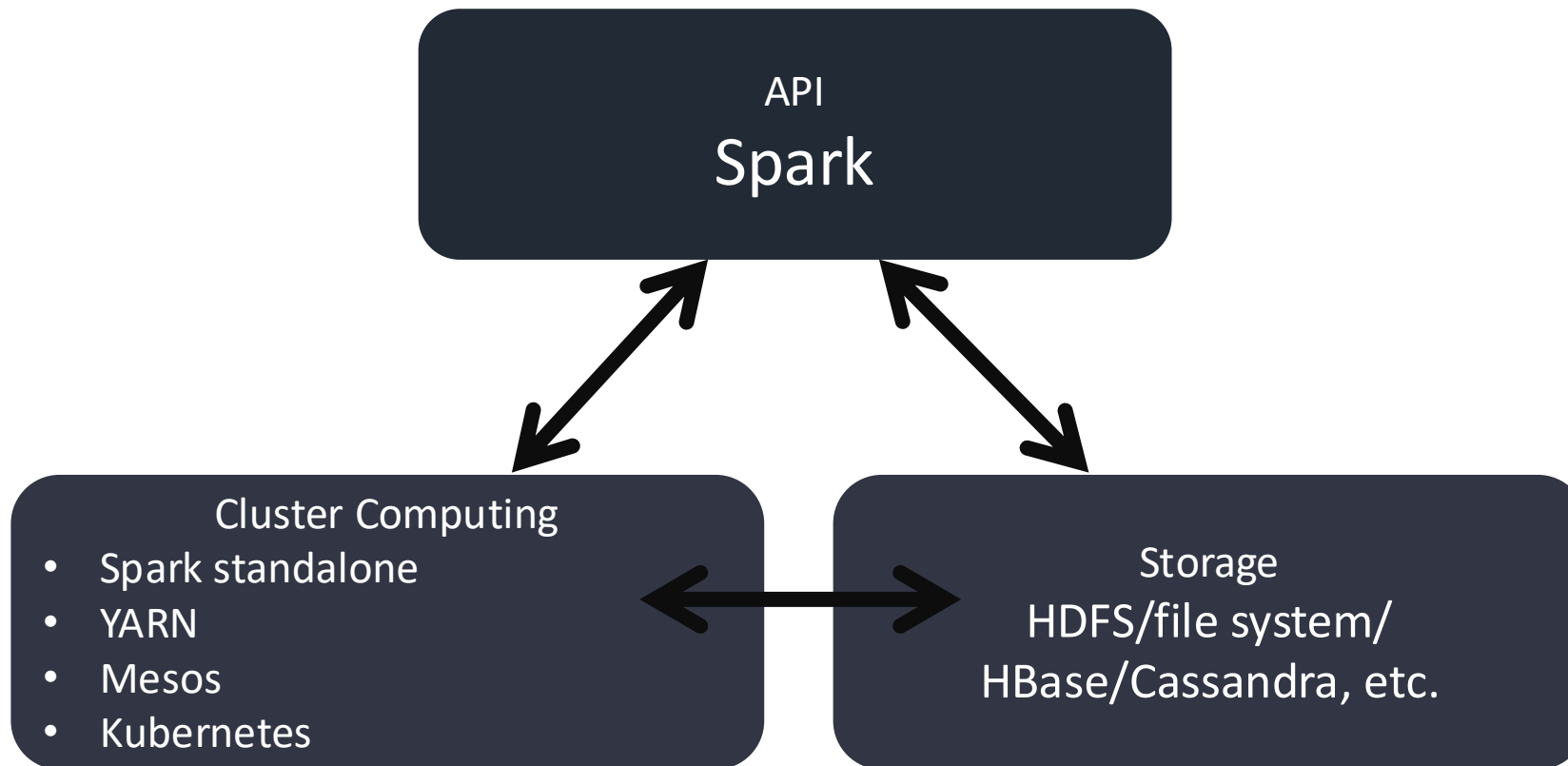


Unit 2. Scalable Computing Paradigm

Introduction to In-Memory Distributed Computing Framework Apache Spark



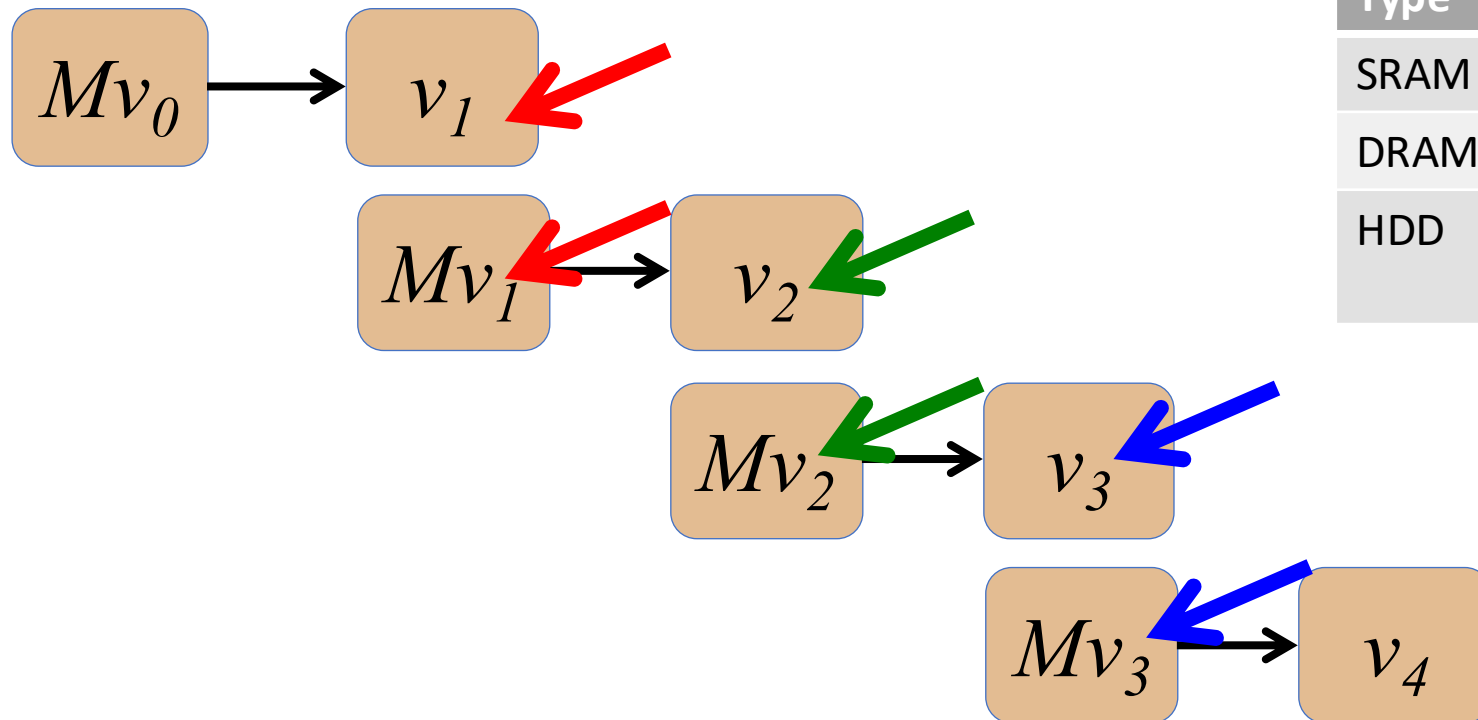
Distributed data processing framework





Why In-Memory Computing?

- Data reuse is common in many iterative machine learning and graph algorithms
 - e.g., AI model training, PageRank, K-means clustering, and logistic regression
- Traditional Big Data frameworks are throttled by disk I/O bottlenecks



Type	Access Time	Rate
SRAM	10 nanoseconds	1
DRAM	50-150 nanoseconds	x 5-15
HDD	9-15 milliseconds	x 900 ~ 1,500



A unified stack

- Spark contains multiple closely integrated components
- Spark core
 - Computational engine
 - Scheduling, distributing, and monitoring applications
- Spark Streaming
 - Processes live streams of data
- MLlib
 - Machine learning functionality
 - ML algorithms (classification, regression, clustering and collaborative filtering)
 - Model evaluation



A unified stack

- GraphX
 - Library for manipulating graphs
 - Performs graph-parallel computations
 - Extends the Spark RDD API
- Cluster Managers
 - Spark can run over a variety of cluster managers
 - Hadoop YARN, Apache Mesos, and Spark built-in cluster manager (Standalone scheduler)

Advantage: The unified stack can minimize data migrations across sub-frameworks

Disadvantage: Some sub-frameworks do not provide the best performance



Questions?